IE3122 – Network Security

Week 1: Module & Lecturer Introduction

Faculty of Computing — BSc (Hons) in Information Technology

Today's Agenda

- Module overview — aim, learning outcomes, delivery structure
- Assessment structure and grading breakdown
- Weekly topic roadmap (Weeks 1–14)
- Meet your lecturer
- Ground rules and how the module runs
- What's next: Networking Revision (this session) & Week 2

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Module Overview

What this module is about, and how it's structured

MODULE AIM

Why This Module Exists

Enable you to identify security vulnerabilities in a network and recommend suitable mitigation techniques — and to evaluate a company's networking requirements and infrastructure to recommend security solutions that protect its network and communications.

Level	Credits	Delivery
Year 3, Semester 1	4 Credits	Lectures 2h + Tutorials 1h + Labs 2h / week

Module Learning Outcomes

LO1	Describe common network vulnerabilities and attacks
LO2	Evaluate mitigation techniques for vulnerabilities and make recommendations
LO3	Analyse a company's network/communication requirements and communicate security requirements
LO4	Evaluate modern security technologies (Firewalls, AAA, IPS/IDS, VPN, Cryptography)
LO5	Configure security technologies (ACL, VPN, Switch/Router Security, Zone-Based Firewalls)

Assessment

How You'll Be Assessed

Component	Weight	Learning Outcomes	Format
Final Exam	50%	LO1–LO5	MCQ + structured essay
Mid-Semester Test	20%	LO1–LO2	MCQ only
Group Assignment	30%	LO2–LO5	Scenario-based report + presentation

- Weekly practical exercises (labs) run throughout the semester and build the hands-on configuration skills tested indirectly via LO5.
- The group assignment is scenario-based: you'll analyse a company's network and recommend security solutions — directly mirroring LO3–LO5.
- The final exam mixes MCQ (recall/LO1–LO2) with structured essay questions (evaluation/analysis, LO3–LO5).

Weekly Roadmap

Weekly Topics — Weeks 1–7

Wk	Topic	Focus
1	Module Intro + Networking Revision	Orientation — layered stack, protocols
2	Networking & Security Revision	Routers/switches, CIA Triad, threat analysis
3	Network Attacks & Malware	Common attacks, vulnerabilities, malware types
4–5	Layer-Specific Attacks	Layer 2, 3 & 4 attacks (2 lectures)
6	Network Defense	Security policy, BCP/DR, malware mitigation
7	Firewalls	Types, stateful inspection, zone-based, placement

Weekly Topics — Weeks 8–14

Wk	Topic	Focus
8	AAA	Server/device-based auth, RADIUS, TACACS+
9	VPN	VPN types, tunneling, SSL VPN vs IPSec
10	IPS and IDS	Functionality, placement, types
11	Cryptography*	Symmetric/asymmetric, block/stream ciphers
12	Digital Certificates*	Hash functions, PKI, non-repudiation
13	Wireless Security	WiFi security, WEP/WPA protocols
14	Revision / Assignment Vivas	Buffer week — no new lecture content

Weekly practicals (20 lab hours) run throughout — switch/router security, ACL, AAA, VPN, zone-based firewalls.

Recommended Texts



CCNA Security 210-260 Official Cert Guide, 1st Ed.

Omar Santos, John Stuppi

Cisco Press



Network Security: Private Communication in a Public World, 3rd Ed.

Charlie Kaufman, Radia Perlman, Mike Speciner, Ray Perlner

Prentice Hall

Meet Your Lecturer

Prof. Harinda Fernando



- **Professor, Dept. of Computer Systems Engineering**

- 18 years in tertiary education across Australia and Sri Lanka
- Core expertise: Cybersecurity, Networking, and Machine Learning
- PhD in Computing, Deakin University (2012)

Current Roles & Research

Current Roles

- Full Professor
- Cluster Lead — Information Assurance & Security Research Cluster
- Program Coordinator, MSc in Cybersecurity

Research Domains

- Cybersecurity
- Networking
- Internet of Things (IoT)
- Applied AI

Recognitions & Supervision

200+
Undergraduate dissertations supervised

30+
Postgraduate dissertations supervised

FHEA
Fellow of the Higher Education Academy

SMIEEE
Senior Member, IEEE

Getting in Touch

- **Email**
- harinda.f@sliit.lk

- **Best way to reach me**
- Email first for anything not urgent; consultation slot details will be shared on the course page.

- **Please include**
- Your student ID and IE3122 in the subject line for faster routing.

Getting Started

How This Module Runs

- **Attendance**
- Lectures, tutorials, and labs are all part of the learning cycle — labs build the configuration skills examined under LO5.

- **Submissions**
- Deadlines for the group assignment will be posted on the course page; late policy follows faculty regulations.

- **Communication**
- Course announcements and materials go through the official course page; email for individual queries.

- **Academic Integrity**
- Group assignment work must be your own team's analysis — standard SLIIT academic integrity policy applies.

Tools You'll Need

Tool	Purpose	When
Cisco Packet Tracer	Switch/router config, ACLs, VPN, firewall labs	From the lab sessions onward
GNS3 (optional)	More advanced topology simulation	As needed for later labs

- No lab session accompanies this Week 1 lecture — instead, this week's practical is an incident-analysis exercise (see the Week 1 lab guide).
- Get Packet Tracer installed before Week 2's session so we can move straight into hands-on work.

What's Next

- **Right now**
- Networking Revision — a quick refresher of the layered stack and core protocols (Ethernet, IP, DHCP, ARP, TCP, UDP, HTTP) before we build on them.

- **This week's lab**
- Recent network security breach case studies — analyse real incidents and answer scenario questions.

- **Week 2**
- Networking and Security Revision — router/switch functionality, the CIA Triad, and threat analysis (LO1, LO3).

Summary

- IE3122 builds skills across 5 LOs: describing threats, evaluating mitigations, analysing requirements, evaluating security technologies, and configuring them.
- Assessment blends MCQ, structured essay, and a scenario-based group assignment — all tied to specific LOs.
- 14 teaching weeks: this orientation week, 12 core topic weeks, and a revision/viva week.
- Reach out early and often — harinda.f@sliit.lk.

Questions?

harinda.f@sliit.lk — Let's move into the Networking Revision